

Diagnostic Environment Design: Posterior-Conditioned Environment Synthesis for Active Preference Learning

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Abstract—When an adaptive agent (a tutor, social companion, or rehabilitation aide) must infer a person’s behavioral preferences from their actions, it has to decide what interactive context to present next. Instead of asking which existing query to show, the agent asks what environment it should create next. We propose *posterior-conditioned environment synthesis*: rather than selecting from a pre-authored stimulus pool, the agent synthesizes new environments online, conditioning each on the current posterior to discriminate among the most-likely preference profiles. We position this as a domain instantiation of adaptive design optimization (ADO/CAT, §II), extending its EIG-argmax core from hand-curated stimulus banks to a parameterized maze-MDP design space. The principal departure is an annotation-light derivation of per-maze contrast vectors in closed form from the maze MDP, with a one-time 4-scalar calibration against 18 hand-annotated key decision points that amortizes the annotation bottleneck about $40\times$ at our largest archive. On a 625-agent synthetic benchmark across three archive configurations, posterior-conditioned selection improves preference recovery over random selection on two of three archives and is statistically indistinguishable on the third. A within-archive factorial isolates the EIG-argmax step, not quality-diversity curation, as the load-bearing operation; a paired LLM-proposer ablation lands at statistical equivalence on this benchmark. The methodological contribution is to lift active preference learning from query selection into environment design.

I. INTRODUCTION

When an agent infers a person’s behavioral preferences from their actions in interactive environments, it must decide what to do next. An assistive manipulator fitting a new user’s reach-effort tradeoff (cf. 1), a social robot calibrating its proactivity to a user’s exploration preference, or a VR tutor adjusting level difficulty must all select interaction contexts that disambiguate competing latent profiles within the first few minutes of contact, exactly the cold-start regime where pool-based query banks saturate. This is the active-elicitation problem at the heart of Bayesian theory of mind in HRI (2, 3) and active preference learning (4, 5). Most work treats elicitation as stimulus or trajectory selection. This paper asks: *what if the system could synthesize new environments, posterior-conditioned, designed to maximally disambiguate among the most likely preference profiles?*

We instantiate that question on a 2D maze domain with Key Decision Points (KDPs) whose branch choice reveals a coordinate of a 4-D latent preference vector $\theta = (\theta_{\text{exp}}, \theta_{\text{eff}}, \theta_{\text{col}}, \theta_{\text{risk}})$. A pre-computed MAP-Elites archive

supplies candidates in milliseconds per probe, and we derive the per-KDP ϕ procedurally from the maze MDP via a closed-form branch-sum (with a one-time 4-scalar calibration against 18 hand-annotated KDPs) that amortizes rather than eliminates the annotation bottleneck. We evaluate the mechanism on a synthetic benchmark; physical HRI deployment is future work.

a) Contributions.: (i) *Posterior-conditioned environment synthesis* as a modality in active preference learning, lifting the EIG-argmax core of adaptive design optimization (ADO/CAT, §II) from hand-curated stimulus banks into a parameterized maze-MDP design space. A within-archive 2×2 factorial (§V-F) confirms EIG-ranking as load-bearing while MAP-Elites’ QD curation adds no identifiable value beyond a same-sized random bank; the novelty relative to ADO/CAT is the maze-MDP design space and the closed-form ϕ derivation (contribution ii), not the EIG-argmax step itself. (ii) *Annotation-light ϕ -derivation* via closed-form branch-sum on the maze MDP, with one-time 4-scalar calibration. (iii) *Cross-archive validation* on a stratified-625 synthetic benchmark: posterior-conditioned selection improves θ -recovery vs. Random-template on 2 of 3 archives (family-mismatched headline: $+3.5 \times 10^{-3}$ and $+3.2 \times 10^{-3}$ L2 units, trial-paired Bonferroni- $k=3$ lower bounds $+1.0$ and $+0.6 \times 10^{-3}$); an AS audit supports the $(1-1/e)$ greedy bound. (iv) *Proposer/selector decomposition*: paired LLM-self-pick vs. LLM-EIG on the same cached $K=3$ batch isolates selector marginal value (§V-E).

II. RELATED WORK

a) Bayesian preference learning and active elicitation.: Three established modalities elicit latent preferences from human behaviour. *Inverse reinforcement learning* (6) infers a reward function from optimal demonstrations; *Bayesian inverse planning* (2, 3) models actions as approximately rational under latent goals, extended to interactive CIRL by Hadfield-Menell et al. (7); and *preference comparison* from human feedback (8) fits a reward model from pairwise rankings. Sadigh et al. (4), Bıyık and Sadigh (5) fuse the latter two into active preference learning by selecting trajectory pairs under EIG. All three operate on *behavioural responses to a fixed task*: they assume the environment is given and elicit preferences over actions, demonstrations, or trajectory rankings within it. The closest established methodology to ours is *adaptive design optimization* (9, 10), which selects from a

parameterized stimulus bank by EIG argmax against a cognitive-model posterior; multidimensional CAT (11, 12) extends this to multi-D latent traits with a similar argmax-EIG operation over a precomputed item bank. Our contribution is a fourth modality relative to behaviour-elicitation, and a domain instantiation relative to ADO/CAT: *environments themselves as queries* in a maze-MDP design space, where the choice variable is the synthesized context presented rather than a stimulus drawn from a pre-authored bank, applicable to tabletop robots, VR tutors, and rehabilitation devices that act in environment-design space. The 2×2 factorial in §V-F confirms that on this benchmark the EIG-argmax step (the ADO core) is the load-bearing operation; the maze-MDP ϕ basis with annotation-light derivation is what is novel relative to ADO’s parameterized-stimulus regime. On the design side, the formal machinery is from BOED (13–15); Golovin and Krause (16)’s adaptive-submodularity framework supplies the $(1 - 1/e)$ greedy bound we audit empirically. Our 4-D 625-state grid admits exact EIG enumeration, so we do not invoke the amortized neural EIG estimators of Foster et al. (15); scaling to higher-D latents would replace the grid with their amortization.

b) Feature learning vs. feature derivation.: Bajcsy et al. (1), Bobu et al. (17, 18) (FERL) tackle the hand-annotation bottleneck by expanding the feature basis from human corrections; Bobu et al. (19) systematizes the logistic observation model we use in Eq. (1). We fix a 4-D basis and derive ϕ loadings procedurally from the maze MDP, trading adaptation for scalability to synthesized environments. The legibility constraint in our synthesis pipeline draws from Dragan et al. (20); procedural content generation (21, 22) provides the parallel thread of generating environments optimized for a metric.

c) LLMs as experimental designers.: Liu et al. (23, 24) instantiate the propose-then-select pattern in hyperparameter optimization and materials discovery; LLMs are competitive with but not dominant over classical acquisition. Mazzaccara et al. (25) show that LLM-emitted 20-questions queries have lower EIG than EIG-best selection from the same K -sample batch, motivating our paired *LLM-self-pick* vs. *LLM-EIG* conditions (same cached batch, $K = 3$; self-consistency pattern of 26) that isolate selector value from proposer quality (§V-E).

III. METHOD

The framework has four components: (i) a logistic 4-D grid posterior over user preferences with a strict consistency invariant, (ii) a parameterized maze design space ψ and a closed-form branch-sum ϕ -derivation pipeline that requires no per-maze manual annotation after a one-time calibration, (iii) a MAP-Elites archive of generated mazes built offline with a hard legibility floor, and (iv) an online posterior-conditioned selection rule (single archive lookup). Component (i) is unchanged from prior pool-based design (9, 10); components (ii)–(iv) are the contributions of this paper.

A. Latent preference model and observation likelihood

A maze environment E is a rectangular grid with start / exit cells, optional items and risk zones, and a set $K(E)$

of Key Decision Points (KDPs). Each KDP is a local binary branch with two cell sets branch_a and branch_b ; the player’s choice $a_k \in \{0, 1\}$ records whether they took the “interest branch.” Each KDP carries a signed, L2-normalized contrast vector $\phi_k(E) \in \mathbb{R}^4$ whose entries load on a 4-D preference basis $\theta = (\theta_{\text{exp}}, \theta_{\text{eff}}, \theta_{\text{col}}, \theta_{\text{risk}})$ grounded in directed exploration (27), effort discounting (28), foraging value (29), and risk attitude (30). The latent space is the 4-D grid $\Theta = \{0, 0.25, 0.5, 0.75, 1.0\}^4$ with $5^4 = 625$ states.

The observation model at each KDP is the standard logistic:

$$P(a_k = 1 \mid \theta, E) = \sigma(\theta^\top \phi_k(E) + b_k), \quad (1)$$

where $\sigma(x) = 1/(1 + e^{-x})$ and the branch bias $b_k = 0$ throughout this paper. The trajectory likelihood factorizes over encountered KDPs under conditional independence given θ , an assumption the validator operationalizes via mutual-reachability checks.

a) Consistency invariant.: The same logistic likelihood is used for (i) posterior update, (ii) held-out predictive probability, and (iii) environment scoring via expected information gain. A SHA-256 `phi_hash` fingerprints ϕ at server boot to detect drift.

B. Parameterized design space ψ

Templates are valid-by-construction in the structurally non-differentiable axes (start/exit reachability, branch-cell adjacency, adversarial reachability of every KDP under any commitment to other KDPs) and parameterize only the geometric knobs that affect ϕ -coverage. A configuration ψ specifies the template family, grid dimensions, KDP count K , and per-slot tuples (trunk_offset, spur_axis, spur_depth, spur_width, interest_branch). Three template families ship in this version (Linear, Loop, and Branching, illustrated in Fig. 1); `g3_loop_*` fixtures share the loop family tag but use a regular-grid stress topology rather than a true perimeter loop. ψ -space cardinality is $\sim 10^7$ at this discretization, of which $\sim 91\%$ produces validator-passing mazes. The narrow bounds are intentional: a template grammar is essential because the validator’s adversarial reachability check forecloses most free-form geometries.

C. Annotation-light ϕ derivation: branch-sum on the maze MDP

Hand-annotated ϕ require introspective judgments (“how much does this branch reveal about exploration vs. effort?”), the same bottleneck FERL (17, 18) addresses by learning features from corrections. We instead derive ϕ_k in closed form from the maze MDP. For each KDP, per-cell features $\phi^{\text{cell}}(c) \in \mathbb{R}^4$ encode whether cell c is a decorative-room cell, an item, a risk-zone cell, and its unit efficiency cost. We sum these per-branch and define ϕ_k as the L2-normalized signed difference between branches; the interest branch sets the sign convention. A 4-vector of per-axis scales calibrates relative magnitudes (uniform vs. 4-scalar Nelder–Mead).

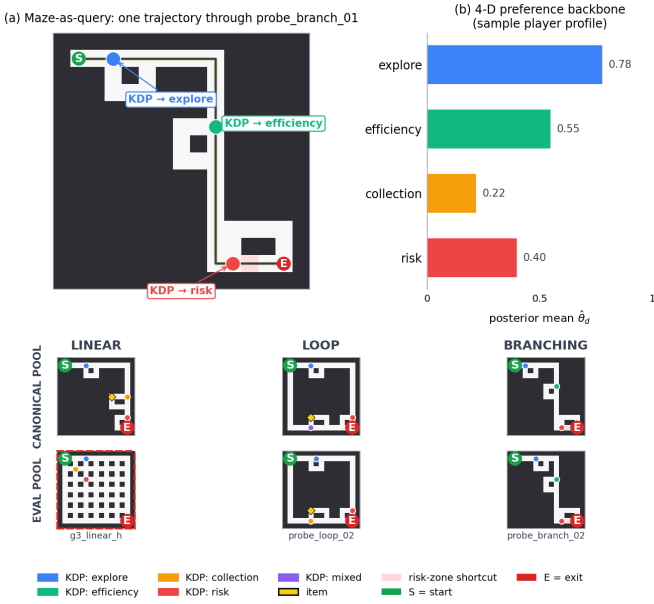


Fig. 1. Maze-as-query primitive. KDPs (markers, colored by axis: blue/green/amber/red for explore/efficiency/collection/risk) are local branch points whose choice signals one preference axis. Top: an illustrative trajectory through `probe_branch_01` with KDP-axis labels, plus the inferred 4-D backbone (posterior means $\hat{\theta}_d$). Bottom: the maze pool, including the `g3_*_h` held-out fixtures (red border).

a) Validation and scale choice.: On 18 hand-annotated KDPs, branch-sum achieves median cosine 0.702 (uniform) vs. 0.923 (calibrated); uniform collapses collection/risk to magnitude ≤ 0.24 , yielding a degenerate archive. We use calibrated scales; other knobs (legibility floor, K , Δ_{eq}) are fixed a priori, not tuned. The v1.2 archive’s $263 \times 3 = 789$ KDPs grounded in 18 anchors give $\sim 44 \times$ amortization.

D. Online selection: EIG and adaptive submodularity

Given the current posterior b_t over Θ , the selection rule picks $E_{t+1} = \arg \max_{E \in \mathcal{A}} \text{EIG}(E \mid b_t)$ over the candidate set $\mathcal{A}$ (excluding within-block repeats), where the expected information gain is

$$\text{EIG}(E \mid b_t) = \mathbb{E}_{\mathbf{a} \sim P(\cdot \mid b_t, E)} [H(\boldsymbol{\theta} \mid b_t) - H(\boldsymbol{\theta} \mid \mathbf{a}, E, b_t)]. \quad (2)$$

EIG is computed exactly by enumerating the 2^K outcome joint and marginalizing over the 625-state grid, tractable at $K \leq 4$.

Adaptive submodularity (AS) (16) provides a $(1-1/e)$ greedy bound under in-expectation submodularity, audited empirically in §V-D. (An earlier $\text{Sep}(E)$ regularizer, a degenerate $\eta=0$ case of D-optimality, was dropped: no decision-theoretic content beyond EIG.)

E. Candidate provenance: archive and LLM proposer

We supply the candidate set $\mathcal{A}$ from three sources, each scored by the same EIG: a 12-maze hand-annotated pool (Pool-DED), a MAP-Elites archive (31–33), and an LLM proposer (23, 34). The archive is built offline by sampling ψ , keying cells by a 5-D descriptor (per-axis ϕ -magnitude

$\times \log \text{Sep}$), and keeping the EIG-maximizing elite per cell subject to a legibility floor; v0/v1.1/v1.2 saturate at 46/134/263 cells. Online lookup is $O(|\mathcal{A}|)$ (~ 5 ms at 263 cells). The LLM proposer serializes b_t and prompts `claude-haiku-4-5` for $K=3$ candidate ψ -vectors per turn (26); two paired conditions share the cached batch (§V-E).

IV. IDENTIFIABILITY AND RECOVERY SCOPE

A. What is being recovered

The 4-D latent $\boldsymbol{\theta} = (\theta_{\text{exp}}, \theta_{\text{eff}}, \theta_{\text{col}}, \theta_{\text{risk}})$ is discretized on a 5-level grid (625 joint states). With turn budgets of 2, 5, or 8 probes per block (3 KDPs per maze, up to 24 binary outcomes per agent), we report two recovery quantities: pointwise L2 distance between posterior mean and true $\boldsymbol{\theta}$ (θ_{ℓ_2}), and posterior-predictive log-likelihood per held-out KDP (LL/KDP). These are not equivalent. The forward map $\boldsymbol{\theta} \mapsto P(a \mid \boldsymbol{\theta}, E)$ is non-injective on the 5-level grid: pairs of $\boldsymbol{\theta}$ that produce identical outcome distributions on the encountered KDPs are indistinguishable post-hoc, so pointwise $\boldsymbol{\theta}$ -recovery is strictly harder than predictive accuracy on the same KDPs.

B. Identifiability via L2 normalization and zero bias

Rationality temperature β_T and the magnitude $\|\boldsymbol{\theta}\|$ are jointly unidentifiable from binary choices, a standard inverse-RL identifiability gap (35). We close it by L2-normalizing ϕ to unit norm and fixing $b_k = 0$ for all KDPs. This pins $\beta_T = 1$ and reduces the recoverable quantity to the *relative* preferences along the 4-axis basis, on the closed cube $[0, 1]^4$. The grid discretization $\Theta = \{0, 0.25, 0.5, 0.75, 1.0\}^4$ then defines a finite posterior support of $5^4 = 625$ states, with uniform prior $P_0(\boldsymbol{\theta}) = 1/625$.

C. Recovery scope at each T

Under the uniform prior the per-dim RMSE floor is ~ 0.29 (variance of uniform on $[0, 1]$). The archive’s held-out evaluation finds per-agent θ_{ℓ_2} clustered around 0.62–0.68 across all turn budgets, well above the floor: ≤ 24 binary observations with per-KDP Fisher information $\leq 1/4$ at $\|\phi\| = 1$ (35 Lemma 1) bound the aggregate Fisher budget at ~ 6 nats, insufficient to concentrate the 625-state posterior to single-cell precision.

The paper’s contribution is therefore not pointwise $\boldsymbol{\theta}$ -recovery in absolute terms (which is bounded by the $T \leq 8$ Fisher budget) but the *relative* comparison: posterior-conditioned MAP-Elites selection improves θ_{ℓ_2} over uniform random sampling (Random-template) on 2 of 3 archive configurations (§V-B); a Random-archive ablation isolates the within-archive selector contribution and identifies it as the load-bearing component on the largest archive. The improvement is small in absolute magnitude — a few percent of the per-agent θ_{ℓ_2} — but is statistically robust (§V-B) and directly cashes out the $(1-1/e)$ adaptive-greedy guarantee empirically established in §V-D.

D. Predictive accuracy: where the signal lives

Held-out predictive LL/KDP is where the gap between MAP-Elites and Pool-DED is most visible (§V-C). A posterior that barely moves off uniform can still produce slightly biased predictive probabilities on held-out KDPs, and bias-adjusted predictive LL accumulates these small biases across the 9 held-out KDPs per agent. The absolute LL/KDP gain is $\sim 10^{-4}$ nats per held-out KDP, far below the noise floor of any single trial but resolvable at $n = 15,000$ paired diffs across 8 seeds.

V. EXPERIMENTS

TABLE I

SELECTION CONDITIONS IN §V: SHARED POSTERIOR, SIMULATOR, AND HELD-OUT PROTOCOL; DIFFER ONLY IN CANDIDATE PROVENANCE AND SELECTION RULE. $|\mathcal{A}|$ APPROXIMATE (WITHIN-BLOCK UNIQUENESS RETRIES REDUCE THE EFFECTIVE SET).

Condition	Candidate source	$ \mathcal{A} $	Selection rule
Pool-DED	12 hand-authored	12	arg max EIG
Random-template	uniform ψ -sample	1	random (baseline)
Random-archive	archive (uniform)	46/134/263	random (no EIG)
MAP-Elites v0	archive (linear)	46	arg max EIG
MAP-Elites v1.1	archive (widened ψ)	134	arg max EIG
MAP-Elites v1.2	archive (branching)	263	arg max EIG
Bank-EIG	random ψ -bank (263)	263	arg max EIG
LLM-Haiku-self	LLM ($K=3$)	3	LLM first-listed
LLM-Haiku-EIG	LLM ($K=3$)	3	arg max EIG

A. Protocol

For every θ on the $5^4 = 625$ grid we run one trial per condition: reset posterior to uniform; for $t = 1 \dots T$ the condition selects a maze, the simulator draws outcomes by (1), and the posterior updates. Held-out predictive LL is evaluated on 3 hand-authored mazes spanning linear, loop, and branching families. Stage 1 uses $T \in \{2, 5, 8\} \times 8$ seeds $\times 625$ agents = 15 000 paired diffs per archive; Stage 2 uses $T = 8 \times 8$ seeds $\times 100$ agents = 800 paired diffs.

a) *CIs and corrections.*: Trial-paired CIs use $B=10,000$ percentile bootstrap over the agent $\times$ seed $\times$ T paired-diff rows; seed-clustered CIs are cluster-bootstrap over the 8 seeds. Bonferroni- k corrections are applied where stated, with $k=3$ for the three ψ -archive comparisons and $k=4$ for the four 2×2 factorial contrasts.

B. Headline: posterior conditioning improves θ -recovery across archives

The most robust cross-archive finding is on ℓ_2 recovery error $\|\hat{\theta} - \theta_{\text{true}}\|_2$, paired against Random-template (Tab. II):

v0 and v1.2 clear Bonferroni- $k=3$ on the top block of Tab. II; v1.1's CI includes zero (traceable to ψ -widening orthogonal to the held-out ϕ distribution). **Primary claim:** MAP-Elites improves θ -recovery over Random-template on 2 of 3 archives, magnitudes $1.0\text{--}3.5 \times 10^{-3}$ L2 units — small but consistent with the AS diminishing-margin envelope (§V-D).

TABLE II

θ -RECOVERY ℓ_2 IMPROVEMENT, PER ARCHIVE ($n=15,000$ PAIRED DIFFS EACH, VALUES $\times 10^{-3}$; SIGN $+$ = ROW RECOVERS θ MORE ACCURATELY THAN COLUMN; CIS TRIAL-PAIRED 95%, SEED-CLUSTERED AND BONFERRONI- $k=3$ IN §V-B PROSE). BOTTOM BLOCK ADDS RANDOM-ARCHIVE TO DECOMPOSE ARCHIVE CURATION FROM WITHIN-ARCHIVE SELECTOR.

contrast	cells	mean	95% CI	z
<i>Total gain vs Random-template (archive + selector combined):</i>				
v0 (linear, fixed)	46	+3.5	[+1.4, +5.5]	+3.33
v1.1 (linear, widened)	134	+1.0	[−1.0, +2.9]	+0.95
v1.2 (linear+branching)	263	+3.2	[+1.0, +5.3]	+2.93
<i>Decomposition (separate run, same protocol):</i>				
EIG selector (MAP—Rand-arch) v0	46	+1.8	[−0.1, +3.7]	+1.85
Archive curation v0	46	+1.2	[−0.8, +3.2]	+1.18
EIG selector v1.1	134	+1.3	[−0.6, +3.3]	+1.34
Archive curation v1.1	134	+0.2	[−1.7, +2.2]	+0.22
EIG selector v1.2	263	+3.6	[+1.5, +5.7]	+3.36
Archive curation v1.2	263	−0.4	[−2.5, +1.6]	−0.40

a) Decomposition: curation vs. within-archive selector:

A *Random-archive* baseline (uniform sample from the same archive, no EIG) separates archive curation from EIG-selector value (Tab. II, bottom). On v1.2 the selector clears trial-paired Bonferroni- $k=3$ but straddles seed-clustered ($[-0.7, +7.0] \times 10^{-3}$); curation alone clears on no archive.

Take-home: v1.2's load-bearing component is the within-archive EIG selector, not archive curation; v0 has neither piece clearing Bonferroni alone but their additive total does; v1.1 is null throughout. The pattern is consistent with the AS diminishing-margin envelope (§V-D): the selector becomes identifiable only when the archive spreads per-batch EIG above the envelope, which v1.2 (the largest archive) does and v1.1 plausibly does not.

Cross-archive headline (8 seeds $\times$ 3 T -values $\times$ 625 agents per archive)

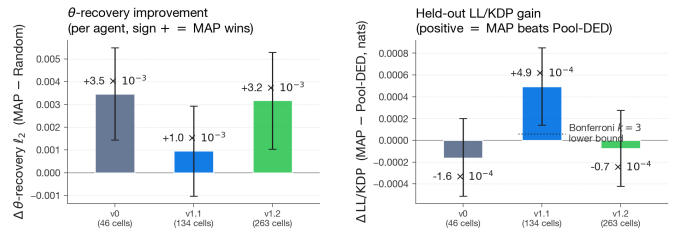


Fig. 2. Cross-archive headline. v0 (46 cells, linear-only), v1.1 (134, widened), v1.2 (263, linear+branching); $n=15,000$ paired diffs each. (Left) θ -recovery ℓ_2 improvement of MAP-Elites over Random-template; v0 and v1.2 95% CIs exclude 0 under Bonferroni- $k=3$. (Right) held-out LL/KDP gain over Pool-DED; only v1.1 has a positive CI (dotted: Bonf- $k=3$ lower bound).

C. Held-out LL: archive-evaluation alignment matters

Held-out LL/KDP gain of MAP-Elites over Pool-DED, per archive ($n=15,000$ paired diffs each, all values $\times 10^{-4}$ nats): v0 -1.6 ($[-5.1, +2.0]$); v1.1 $+4.9$ ($[+1.3, +8.5]$, Bonferroni- $k=3$ $[+0.6, +9.3]$); v1.2 -0.7 ($[-4.2, +2.8]$). The LL/KDP picture is inferential-frame-conditional. Under trial-paired CIs only v1.1 clears Bonferroni- $k=3$ (LB $+0.6 \times 10^{-4}$ nats); v0

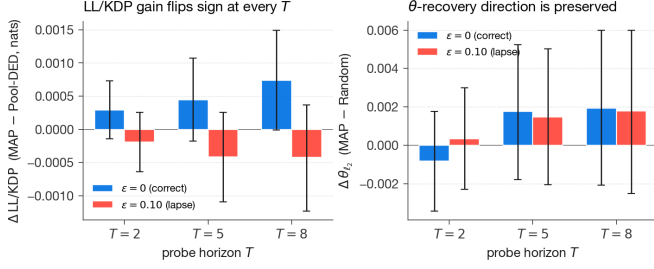


Fig. 3. v1.1 brittleness under lapse misspecification. (Left) held-out LL/KDP gain of MAP-Elites over Pool-DED at $T \in \{2, 5, 8\}$, $\epsilon=0$ vs. flipped-outcome lapse $\epsilon=0.10$: on-model wins $(+2.9, +4.5, +7.4 \times 10^{-4})$ flip to misspec losses $(-1.9, -4.1, -4.2 \times 10^{-4})$ nats. (Right) Same for θ -recovery, preserved across lapse settings.

TABLE III

ADAPTIVE SUBMODULARITY AUDIT (v0 ARCHIVE, 46 CELLS, $K=3$, $n=10,000$ RANDOM TUPLES). THE DIMINISHING-MARGIN ENVELOPE ($p_{50}=1.35 \times 10^{-3}$ NATS) BOUNDS THE PER-STEP EIG HEADROOM THAT THE LLM PROPOSER IN §V-E COULD NOT EXCEED.

Quantity	Value
In-expectation violations	0/10,000
Pointwise violations	49.9%
Mean pointwise violation mass	0.20
Worst single-tuple inflation	6.34×10^{-2}
Diminishing margin (median)	1.35×10^{-3} nats
Diminishing margin (p95)	2.61×10^{-3} nats

saturates and v1.2 is negative on the family-mismatched 3-maze held-out, which §V-F’s in-domain re-evaluation attributes to held-out distribution shift rather than selector over-fitting. The more conservative seed-clustered 95% CI ($n=8$ seeds) includes zero for v1.1: the LL/KDP win is established within-archive but not between seeds. We report both flavors throughout, treating trial-paired as effect-size estimate and seed-clustered as conservative bound. At $\epsilon = 0.10$ lapse, LL/KDP inverts at every T while θ -recovery direction holds, exactly the brittleness Bobu et al. (36) predicts (Fig. 3).

D. Adaptive submodularity holds in expectation

We audit $n = 10,000$ random tuples $(b, E_{\text{obs}}, E_{\text{test}})$ on v0 (46 cells, $K=3$) for the in-expectation inequality $\text{EIG}(E_{\text{test}} | b) \geq \mathbb{E}_y[\text{EIG}(E_{\text{test}} | b | (E_{\text{obs}}, y))]$; results in Table III. Submodularity holds in expectation, invoking the $(1-1/e)$ regret bound of Golovin and Krause (16), and the diminishing-margin envelope is small so greedy is near-optimal. Pointwise non-monotonicity is real but unexploitable: a 1-step belief-MDP lookahead lost to greedy myopic by -7.7×10^{-3} nats on a pre-registered pilot, dominated by the diminishing-margin envelope.

E. EIG selector vs LLM self-rank: a paired isolation

Both LLM-Haiku-self-pick (first-listed) and LLM-Haiku-EIG (argmax-EIG) consume the same $K=3$

claude-haiku-4-5 batch per turn (§III-E), isolating the selector with proposer held constant.

a) *Headline (selector isolation).*: LLM-Haiku-EIG – LLM-Haiku-self-pick on held-out LL/KDP at $T=8$ is -7.8×10^{-4} nats/KDP (95% CI $[-2.5, +0.96] \times 10^{-3}$, $n=800$). With equivalence margin $\Delta_{\text{eq}}=5 \times 10^{-3}$ set a priori as $\sim 3 \times$ the AS diminishing-margin envelope, TOST returns **EQUIVALENT** (90% CI $[-2.2, +0.67] \times 10^{-3}$). The EIG selector and the LLM’s natural-language EIG heuristic are statistically equivalent at $K=3$ on the Haiku-class proposer.

b) *Selector decomposition.*: All four pairwise LL/KDP comparisons among LLM-self-pick, LLM-EIG, MAP-Elites, and Pool-DED at $T=8$ yield CIs including zero (e.g., LLM-EIG – MAP-Elites -1.2×10^{-3} ; LLM-self-pick – MAP-Elites -4.7×10^{-4}); the offline-optimized archive is statistically indistinguishable from LLM proposals scored by the same selector.

c) *Diagnostics audit.*: Per-batch EIG spread (mean $\approx 1.0 \times 10^{-3}$ nats over 6,400 turns) sits below the AS envelope (Table III) at every $T \in \{2, 5, 8\}$, isolating the null at the proposer.

F. Cautionary ablation: archive expressiveness has diminishing returns

v1.2’s branching template strengthens θ -recovery while reversing the LL/KDP gain. The fork-KDP’s asymmetric ϕ -vectors sharpen the posterior in directions concentrated within the archive distribution; the family-mismatched held-out confounds this with distribution shift.

a) *In-domain held-out (12 v1.2 archive cells).*: Re-running v1.2 with held-out drawn from the same archive distribution (a within-distribution generalization probe, not strict held-out) shrinks the EIG-selector LL gap from clearly negative on family-mismatched (-5.3×10^{-4}) to null in-domain (-2.1×10^{-4}). Curation moves the other way, from null to directionally negative. *theta-recovery strengthens*: the v1.2 EIG-selector gain is $+3.1 \times 10^{-3}$ with seed-clustered Bonferroni- $k=3$ $[+0.30, +5.4] \times 10^{-3}$ clearing zero, against the family-mismatched straddle $[-0.7, +7.0] \times 10^{-3}$.

The original “selector over-fits” framing was partly an artifact of held-out family-mismatch. On in-domain held-out the selector is null on LL and archive concentration itself contributes the small residual negative, while θ -recovery transfers cleanly.

b) *Cross-archive in-domain replication (v0 and v1.1).*: Parallel in-domain runs on v0 (46 cells) and v1.1 (134 cells), each with $n=12$ held-out, surface an *archive-size-dependent* split rather than a single generalizable claim. On v0 the load-bearing contribution shifts from selector to curation (selector θ_{ℓ_2} collapses to null; curation becomes seed-Bonferroni- $k=3$ identifiable). v0’s selector LL gap is robustly negative across both held-outs, not a family-mismatch artifact. v1.1 is null on every contrast in both regimes. v1.2 inverts the family-mismatched verdict on each metric: selector θ_{ℓ_2} moves from straddle to clear; selector LL/KDP from clearly negative to null; curation LL turns directionally negative.

c) 2×2 factorial: *EIG ranking is load-bearing, MAP-Elites curation is null.*: A further ablation isolates the curation step from the EIG ranking step on the v1.2 archive. We add a fourth condition, **Bank-EIG**: argmax-EIG over a fixed bank of 263 random ψ -samples (deterministic seed, validator-passing, no QD curation, no overlap with held-out). This completes the 2×2 factorial of *curation* $\times$ *EIG ranking*.

TABLE IV

2×2 FACTORIAL ON v1.2 IN-DOMAIN ($n=15,000$ PAIRED DIFFS; 95% CIs TRIAL-PAIRED / SEED-CLUSTERED; VALUES $\times 10^{-3}$ ON θ_{ℓ_2}). EIG-RANKING CONTRASTS CLEAR ZERO ON BOTH FLAVORS; CURATION CONTRASTS STRADDLE.

contrast	mean	trial-paired	seed-clustered
<i>EIG ranking (load-bearing):</i>			
MAP – Random-arch	+3.12	[+1.05, +5.23]	[+0.97, +5.10]
Bank-EIG – Random-arch	+3.73	[+1.59, +5.86]	[+1.17, +6.29]
Bank-EIG – Random-templ	+3.80	[+1.73, +5.96]	[+1.92, +6.01]
<i>Curation (null):</i>			
MAP – Bank-EIG	−0.61	[−2.80, +1.56]	[−3.28, +2.40]

Bonferroni- $k=4$ on the four factorial contrasts: smallest EIG-on lower bound $+0.32 \times 10^{-3}$ (MAP – Random-arch seed-clustered); the curation contrast widens to $[-3.47, +2.12] \times 10^{-3}$ and straddles (§VI-C). On LL/KDP, curation is similarly null ($+0.48 \times 10^{-4}$, $[-2.15, +3.03]$ trial-paired). *QD-curation adds no identifiable contribution beyond EIG argmax over a same-sized random bank.* The v1.2 contribution scopes to the EIG-ranking operation over a maze-MDP design space, with MAP-Elites reduced to fast-lookup infrastructure.

VI. DISCUSSION

A. What the data say, conservatively

The EIG-ranking step is load-bearing on θ -recovery; MAP-Elites quality-diversity curation, holding EIG fixed, contributes no identifiable additional value. Three claims, each with a disclosable scope:

a) *EIG ranking, not quality-diversity (QD) curation, drives θ -recovery.*: The 2×2 factorial on v1.2 in-domain (Tab. IV) shows all three EIG-on contrasts clear zero on θ_{ℓ_2} , while curation alone (MAP – Bank-EIG) straddles. v1.2 is the only archive where the EIG step survives Bonferroni in-domain (v0 near-floor, v1.1 null).

b) *LL/KDP holds only on family-matched evaluation.*: v1.1’s trial-paired Bonferroni win does not survive seed-clustered CIs (§V-C); v0 and v1.2 are negative on family-mismatched LL/KDP, but v1.2’s gap resolves to null in-domain (distribution shift, not selector over-fitting). We treat θ -recovery as the load-bearing metric.

c) *LLM-proposer is the bottleneck, not the selector.*: LLM-self-pick and LLM-EIG land at TOST equivalence at $\pm 5 \times 10^{-3}$ (§V-E): the per-batch EIG spread of 1.0×10^{-3} nats lies below the AS diminishing-margin envelope (1.35×10^{-3} , §V-D), so the selector has nothing to differentiate; the proposer is the rate-limiting step.

B. Pipeline feasibility ($N=6$, non-IRB)

An $N=6$ non-IRB run confirmed the four-block flow runs end-to-end. With $N=3$ per self-report stratum we do not interpret correlations as construct validity, but two maze-pool issues surface for any follow-up powered study: collection KDPs carry an efficiency-cost detour (preventing $\hat{\theta}_{\text{col}}$ separation), and risky-shortcut KDPs are long enough that even risk-averse participants took the shortcut.

C. Limitations and future work

Human evidence is feasibility-only; a larger- N within-subject study is needed for powered between-condition comparisons. **Published held-out is small** ($n=3$ mazes, family-mismatched); $n=12$ in-domain re-evaluation in §V-F resolves an archive-size-dependent contribution split (curation on v0, selector on v1.2, null on v1.1). **Effect sizes are small** ($\sim 5 \times 10^{-4}$ nats LL/KDP); the value is the annotation-light closed-form principle, not magnitude. **Misspecification brittleness**: v1.1 LL/KDP inverts at $\varepsilon=0.10$, exactly as Bobu et al. (36) predicts. **LLM-ablation scope**: fixed $K=3$, single zero-shot proposer; whether the selector gap returns at larger K or stronger models is open. **Multiplicity**: Bonferroni- $k=3$ corrects for three archive comparisons; widening to $k=9$ keeps the v1.2 in-domain selector headline trial-paired ($[+0.25, +6.10] \times 10^{-3}$) but seed-clustered $k=18$ straddles ($[-0.49, +5.99] \times 10^{-3}$). Read the headline as trial-paired identifiability with seed-clustered boundary, not universal clearing.

a) *Future work.*: (1) Pre-registered single-CI analysis matching the archive-by-decomposition contrast surface; (2) LLM-proposer scaling (sweep K , stronger models, reflective prompting, fine-tuning (25)); (3) differentiable QD (33) via the implemented JAX EIG gradient. The environment-as-query primitive extends to assistive-arm calibration, social-robot personalization, and adaptive VR tutoring.

VII. CONCLUSION

We instantiated posterior-conditioned environment synthesis as an alternative to pool selection: parameterized templates pass the validator cheaply, branch-sum ϕ derivation amortizes per-maze annotation via one-time calibration, a MAP-Elites archive supports fast online lookup (a 2×2 factorial finds EIG-argmax over a same-sized random bank achieves the same θ -recovery, so MAP-Elites is infrastructure, not a load-bearing claim), and greedy myopic EIG selection is principled by an AS audit. The defended claim is conservative: θ -recovery improves on 2 of 3 archives (family-mismatched: $+3.5$ and $+3.2 \times 10^{-3}$, Bonferroni- $k=3$ lower bounds $+1.0$ and $+0.6 \times 10^{-3}$); LL/KDP gains are archive-evaluation-alignment-sensitive. The methodological contribution is that interactive environments themselves can be active queries, selected online via EIG-argmax over a maze-MDP design space, with ϕ derived procedurally with one-time calibration, opening a class of adaptive HRI problems where the agent reshapes the interaction.

REFERENCES

- [1] Andrea Bajcsy, Dylan P. Losey, Marcia K. O'Malley, and Anca D. Dragan. Learning from physical human corrections, one feature at a time. In *Proc. ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 2018.
- [2] Chris L. Baker, Julian Jara-Ettinger, Rebecca Saxe, and Joshua B. Tenenbaum. Rational quantitative attribution of beliefs, desires and percepts in human mentalizing. *Nature Human Behaviour*, 1(4):0064, 2017.
- [3] Julian Jara-Ettinger, Hyowon Gweon, Laura E. Schulz, and Joshua B. Tenenbaum. The naïve utility calculus: Computational principles underlying commonsense psychology. *Trends in Cognitive Sciences*, 20(8):589–604, 2016.
- [4] Dorsa Sadigh, Anca D. Dragan, S. Shankar Sastry, and Sanjit A. Seshia. Active preference-based learning of reward functions. In *Proc. Robotics: Science and Systems (RSS)*, 2017.
- [5] Erdem Bıyık and Dorsa Sadigh. Batch active preference-based learning of reward functions. In *Proc. Conference on Robot Learning (CoRL)*, 2018.
- [6] Andrew Y. Ng and Stuart J. Russell. Algorithms for inverse reinforcement learning. In *Proc. ICML*, 2000.
- [7] Dylan Hadfield-Menell, Stuart J. Russell, Pieter Abbeel, and Anca Dragan. Cooperative inverse reinforcement learning. In *Proc. NeurIPS*, 2016.
- [8] Paul F. Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Proc. NeurIPS*, 2017.
- [9] Daniel R. Cavagnaro, Jay I. Myung, Mark A. Pitt, and Janne V. Kujala. Adaptive design optimization: A mutual information-based approach to model discrimination in cognitive science. *Neural Computation*, 22(4):887–905, 2010.
- [10] Jaeyeong Yang, Mark A. Pitt, Woo-Young Ahn, and Jay I. Myung. ADOPy: A Python package for adaptive design optimization. *Behavior Research Methods*, 53(2):874–897, 2021.
- [11] Daniel O. Segall. Multidimensional adaptive testing. *Psychometrika*, 61(2):331–354, 1996.
- [12] Joris Mulder and Wim J. van der Linden. Multidimensional adaptive testing with optimal design criteria for item selection. *Psychometrika*, 74(2):273–296, 2009.
- [13] D. V. Lindley. On a measure of the information provided by an experiment. *The Annals of Mathematical Statistics*, 27(4):986–1005, 1956.
- [14] Kathryn Chaloner and Isabella Verdinelli. Bayesian experimental design: A review. *Statistical Science*, 10(3): 273–304, 1995.
- [15] Adam Foster, Desi R. Ivanova, Ilyas Malik, and Tom Rainforth. Deep adaptive design: Amortizing sequential Bayesian experimental design. In *Proc. ICML*, 2021.
- [16] Daniel Golovin and Andreas Krause. Adaptive submodularity: Theory and applications in active learning and stochastic optimization. *Journal of Artificial Intelligence Research*, 42:427–486, 2011.
- [17] Andreea Bobu, Marius Wiggert, Claire J. Tomlin, and Anca D. Dragan. Feature expansive reward learning: Rethinking human input. In *Proceedings of the 2021 ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 2021. doi: 10.1145/3434073.3444667.
- [18] Andreea Bobu, Marius Wiggert, Claire J. Tomlin, and Anca D. Dragan. Inducing structure in reward learning by learning features. *The International Journal of Robotics Research*, 41(5):497–518, 2022.
- [19] Andreea Bobu, Dexter R. R. Scobee, Jaime F. Fisac, S. Shankar Sastry, and Anca D. Dragan. LESS is more: Rethinking probabilistic models of human behavior. In *Proceedings of the 2020 ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 2020.
- [20] Anca D. Dragan, Kenton C. T. Lee, and Siddhartha S. Srinivasa. Legibility and predictability of robot motion. In *Proceedings of the 2013 ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 2013.
- [21] Georgios N. Yannakakis and Julian Togelius. Experience-driven procedural content generation. *IEEE Transactions on Affective Computing*, 2(3):147–161, 2011.
- [22] Ahmed Khalifa, Philip Bontrager, Sam Earle, and Julian Togelius. PCGRL: Procedural content generation via reinforcement learning. *AIIDE*, 2020.
- [23] Tension Liu, Nicolás Astorga, Nabeel Seedat, and Mihaela van der Schaar. Large language models to enhance Bayesian optimization. In *Proc. ICLR*, 2024.
- [24] Tension Liu et al. LLMs for Bayesian optimization in scientific domains: Are we there yet? In *Findings of EMNLP*, 2025. arXiv:2509.21403.
- [25] Davide Mazzaccara, Alberto Testoni, and Raffaella Bernardi. Learning to ask informative questions: Enhancing LLMs with preference optimization and expected information gain. In *Findings of EMNLP*, 2024.
- [26] Xuezhi Wang, Jason Wei, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *Proc. ICLR*, 2023.
- [27] Robert C. Wilson, Andra Geana, John M. White, Elliot A. Ludvig, and Jonathan D. Cohen. Humans use directed and random exploration to solve the explore-exploit dilemma. *Journal of Experimental Psychology: General*, 143(6): 2074–2081, 2014.
- [28] Wouter Kool and Matthew Botvinick. A labor/leisure tradeoff in cognitive control. *Journal of Experimental Psychology: General*, 143(1):131–141, 2014.
- [29] Nils Kolling, Timothy E. J. Behrens, Rogier B. Mars, and Matthew F. S. Rushworth. Neural mechanisms of foraging. *Science*, 336(6077):95–98, 2012.
- [30] Amos Tversky and Daniel Kahneman. Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty*, 5(4):297–323, 1992.

- [31] Jean-Baptiste Mouret and Jeff Clune. Illuminating search spaces by mapping elites. *arXiv preprint arXiv:1504.04909*, 2015. URL <https://arxiv.org/abs/1504.04909>.
- [32] Antoine Cully, Jeff Clune, Danesh Tarapore, and Jean-Baptiste Mouret. Robots that can adapt like animals. *Nature*, 521(7553):503–507, 2015. doi: 10.1038/nature14422.
- [33] Matthew C. Fontaine and Stefanos Nikolaidis. Differentiable quality diversity. In *Advances in Neural Information Processing Systems 34 (NeurIPS)*, 2021.
- [34] Akash Choudhury, Robert Williamson, Adam Golinski, Yifan Miao, Freddie Bickford Smith, Michael Kirchhof, Andrew Zhang, and Tom Rainforth. BED-LLM: Intelligent information gathering with LLMs and Bayesian experimental design. In *Proc. ICLR*, 2026. arXiv:2508.21184.
- [35] Rohin Shah, Noah Gundotra, Pieter Abbeel, and Anca D. Dragan. On the feasibility of learning, rather than assuming, human biases for reward inference. In *Proc. ICML*, 2019.
- [36] Andreea Bobu, Andrea Bajcsy, Jaime F. Fisac, Sampada Deglurkar, and Anca D. Dragan. Quantifying hypothesis space misspecification in learning from human-robot demonstrations and physical corrections. *arXiv preprint arXiv:2002.00941*, 2020. URL <https://arxiv.org/abs/2002.00941>.